

Global fossil methane emissions constrained by multi-isotopic atmospheric methane histories

Ryo Fujita^{1,2*}, Heather Graven¹, Giulia Zazzeri^{1,3}, Benjamin Hmiel^{4, 5},
Vasilii V. Petrenko⁴, Andrew M. Smith⁶, Sylvia E. Michel⁷, Shinji Morimoto⁸

Affiliations:

¹Department of Physics, Imperial College London, London, United Kingdom

²Meteorological Research Institute, Japan Meteorological Agency, Tsukuba, Japan

³Ricerca sul Sistema Energetico (RSE), Milan, Italy

⁴Department of Earth and Environmental Sciences, University of Rochester, Rochester, NY, USA

⁵Environmental Defense Fund, NY, USA

⁶Centre for Accelerator Science, Australian Nuclear Science and Technology Organisation (ANSTO),
Lucas Heights, Australia

⁷Institute of Arctic and Alpine Research, University of Colorado, Boulder, Colorado, USA

⁸Center for Atmospheric and Oceanic Studies, Graduate School of Science,
Tohoku University, Sendai, Japan

*Corresponding author. Email: ryo.fujita@mri-jma.go.jp

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Isotope definition

The definition of $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ are:

$$\delta^{13}\text{C} = \left(\frac{\left(\frac{^{13}\text{CH}_4}{^{12}\text{CH}_4} \right)_{\text{sample}}}{\left(\frac{^{13}\text{C}}{^{12}\text{C}} \right)_{\text{VPDB}}} - 1 \right) \times 1,000 \quad (\text{‰}), \quad (\text{S1})$$

$$\delta\text{D} = \left(\frac{\left(\frac{\text{CH}_3\text{D}}{\text{CH}_4} \right)_{\text{sample}}}{\left(\frac{\text{D}}{\text{H}} \right)_{\text{VSMOW}}} - 1 \right) \times 1,000 \quad (\text{‰}), \quad (\text{S2})$$

where subscript “*sample*” denotes the sample, and Vienna Pee Dee belemnite (VPDB) represents the international standard for $\delta^{13}\text{C-CH}_4$, with $^{13}\text{C}/^{12}\text{C}$ of $1,123.72 \times 10^{-5}$ (Craig, 1957), and the Vienna standard mean ocean water (VSMOW) for $\delta\text{D-CH}_4$, with D/H of 155.76×10^{-6} (Hagemann et al., 1970).

The definition of $\Delta^{14}\text{C-CH}_4$ is following Lassey et al. (2007) as:

$$\Delta^{14}\text{C} = \left(\frac{A_{\text{SN}}}{A_{\text{abs}}} - 1 \right) \times 1,000 \quad (\text{‰}), \quad (\text{S3})$$

$$A_{\text{SN}} = A_{\text{S}} \left(\frac{0.975}{1 + \delta^{13}\text{C}/1,000} \right)^2 \quad (\text{‰}), \quad (\text{S4})$$

where A_{abs} is the “absolute international standard activity” of $0.2260 \pm 0.0012 \text{ Bq gC}^{-1}$ (Stuiver, 1980), Bq is Becquerel, or disintegrations per second equates to $433.2 \times 10^{-15} \text{ mole}(^{14}\text{C})$, and A_{SN} is the “normalized” sample activity (assumed to be corrected for “shelf” decay between sample collection and ^{14}C analysis) expressed in terms of the measured activity A_{S} (i.e. $^{14}\text{CH}_4/\text{total CH}_4$ of sample) normalized to a $\delta^{13}\text{C-CH}_4$ of -25‰ (see Lassey et al. (2007) for more detail).

Observation targets

We set atmospheric CH_4 , $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and $\Delta^{14}\text{C-CH}_4$ observational target ranges for 51 target years, 1750, 1850, 1900, 1950, 1960, and every year for 1970–2015 (Fig. S1 and Table S1). The mean and uncertainty of targets, for which the values were not prescribed, were linearly interpolated using the adjacent data. The atmospheric target year is assumed to be at 0:00 UTC, January 1st in each year; thus the annual observational targets are interpolated when adjacent annual mean data is available (e.g., target data at 2000.0 is derived from annual mean in 1999 (year = 1999.5) and in 2000 (year = 2000.5)) (Table S1). Since the setting of atmospheric CH_4 is already described in the main text, here we describe how atmospheric $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and $\Delta^{14}\text{C-CH}_4$ targets were constructed in more detail.

Since the spatiotemporal coverages of these observations were limited before 1970s, especially

for $\delta\text{D-CH}_4$ and $\Delta^{14}\text{C-CH}_4$, it is difficult to reconstruct annual-basis global observational targets. Thus we adopted multi-decadal blocks of observational target before 1970. To reconstruct the global mean and uncertainty under the sparse data coverage, we often utilized the Arctic and Antarctic data as the boundary value of global tropospheric mean (see below). Although these approaches should weaken their actual observational constraints, we confirmed such historical targets can still constrain the average CH_4 source and sink parameters, particularly natural source parameters (see section 3.1 and Fig. 4 in the main text).

The $\delta^{13}\text{C-CH}_4$ target ranges over 1750–1900 and $\delta\text{D-CH}_4$ over 1750–1970 are assumed to be time-invariant, which are inferred from Antarctic ice-core and firn-air observational data in Ferretti et al. (2005) and Mischler et al. (2009). The target ranges are set to be wide, $[-50, -48] \text{‰}$ for $\delta^{13}\text{C-CH}_4$ and $[-115, -85] \text{‰}$ for $\delta\text{D-CH}_4$, respectively, considering the limited time and space representativeness of the observation, the inter-hemispheric differences in $\delta^{13}\text{C-CH}_4$ ($\sim 0.6\text{‰}$) and $\delta\text{D-CH}_4$ ($\sim 16\text{‰}$) (Beck et al., 2018), and their potential inter-laboratory measurement differences (Umezawa et al., 2018). For $\delta^{13}\text{C-CH}_4$ over 1900–1970, a secular increasing trend, 0.015‰/year , is estimated from the data in Ferretti et al. (2005), Mischler et al. (2009), and Rice et al. (2016). The linear trend is applied to the respective lower and upper bounds of the target from 1900 every 10 years to 1970. It is known that the atmospheric krypton (Kr) can lead to biases in $\delta^{13}\text{C-CH}_4$ during mass spectrometric measurements, estimated to be about $+0.4\text{‰}$ on Ferretti et al. (2005) and Mischler et al. (2009)’s $\delta^{13}\text{C-CH}_4$ data (Schmitt et al., 2013). Also, Ferretti et al. (2005)’s data have not been corrected for diffusive fractionation in the firn air at Law Dome DE-08 site, estimated to be up to -0.4‰ (the largest during the period of rapid atmospheric CH_4 increase around 1970) (Sapart et al., 2013). Since our historical $\delta^{13}\text{C-CH}_4$ target ranges are specified to be quite wide, allowing a wide range of $\delta^{13}\text{C-CH}_4$ trends over 1750–1970, we assume the potential biases for both Kr interference and diffusive fractionation in Ferretti et al. (2005) and Mischler et al. (2009)’s data are not significant in our analysis.

For 1984–2013, annual mean globally averaged $\delta^{13}\text{C-CH}_4$ data is taken from SI Figure 3 of Schwietzke et al. (2016), where INSTAAR $\delta^{13}\text{C-CH}_4$ data is used for 1999–2013 (Michel et al., 2021) and NIWA $\delta^{13}\text{C-CH}_4$ data is used for 1984–1998 (Schaefer et al., 2016) after unifying to INSTAAR $\delta^{13}\text{C-CH}_4$ scale (see Schwietzke et al. (2016)). Before 1984, the global average is extrapolated until 1978 using northern hemispheric composite $\delta^{13}\text{C-CH}_4$ trend in Rice et al. (2016) for 1978–1983 and the mean difference of the global averages from the northern composite $\delta^{13}\text{C-CH}_4$ for 1984–2009, $0.15 \pm 0.04\text{‰}$ (1 SD). Note that the difference includes both the spatial representative difference and potential inter-laboratory measurement differences (Umezawa et al., 2018).

For $\delta\text{D-CH}_4$, the annual mean globally average is derived from global coverage observational network by Tohoku University (TU) for 2006–2013 (Fujita et al., 2020) and INSTAAR for 2006–2009 (except Black Sea, Romania, for the purpose of background site selection) (White et al., 2016) (see Fig. S1). The procedure for obtaining the global averages and uncertainties is as follows (updated from Turner et al. (2017) for obtaining global averages):

1. Compute annual averages of $\delta\text{D-CH}_4$ at each station using long-term trend to the observed data estimated by a digital-filtering technique (Nakazawa et al., 1997).
2. Group each annual mean record by 4 latitudinal bands: NET (Northern Extra Tropics: 30°N – 90°N), NTRO (Northern Tropics: EQ – 30°N), STRO (Southern Tropics: 30°S – EQ), and SET (Southern Extra Tropics: 90°S – 30°S) and determine the number of sites in each latitude: n_{NET} , n_{NTRO} , n_{STRO} , and, n_{SET} .
3. Randomly select n_{XX} sites from each latitudinal band with replacement.

4. Compute the zonal means in 4 latitudinal bands from the randomly selected n_{XX} sites and derive global average by averaging the 4 zonal means.
5. Repeat 1000 times, yielding 1000 different global averages.
6. Compute the mean global average and standard deviation from the 1000 realizations.

In step 4, a random error of 2‰ (1 SD) is applied when generating the annual mean for each station as the inclusion of measurement uncertainty and spatial representativeness (White et al., 2016; Fujita et al., 2020). The shipboard data in the western Pacific region in Fujita et al. (2020) is combined into 4 latitudinal band (33°N–20°N, 15°N–EQ, 20°S–5°S, and 40°S–25°S), which are grouped into each latitudinal band in step 2 using their mean latitudes. There is no significant difference in global averages calculated from either TU or INSTAAR dataset for 2006–2009; thus we combine their data for 2006–2009 and the global averages from TU dataset for 2010–2013. Before 2005, the global average is extrapolated by averaging the NET and SET trends for 1995–2005 (Rice et al., 2016; Fujita et al., 2020) and incorporating the mean difference between global averages and NET and SET averages for 2006–2013 ($1.5 \pm 0.2\%$). Additionally, the extrapolation includes the NET trend for 1978–1994 (Rice et al., 2016) and the mean difference between global averages and NET for 2006–2013 ($8.9 \pm 0.4\%$). The potential inter-laboratory differences are added to Rice et al. (2016)’s $\delta D-CH_4$ data ($-2.4 \pm 1.6\%$) (Umezawa et al., 2018) to adjust to the TU $\delta D-CH_4$ scale (Fujita et al., 2020).

The observational $\Delta^{14}C-CH_4$ targets for 1750–1970 are constructed using the Greenland and Antarctic ice-core and firn-air data in Hmiel et al. (2020). First, the measured data, whose mean age of air is around 25 years from 1750 and 1900 and around 10 years from 1950, 1960, 1970 are selected. Then, the target range in each target year is determined by using the minimum and maximum among these data—mean ± 2 SD of each data in Table S2 of Hmiel et al. (2020). Note that the target range is specified to be wider if the data number is less than three or the dataset does not contain both northern and southern hemispheric data, to take into account the limited spatiotemporal representativeness in the measured data over the period. Only one measurement at Summit, Greenland, is available in 1750, thus the target range is widened by mean ± 4 SD. No northern hemispheric data are available in 1950, thus the measured data at Summit whose mean age of air is 1966 are also used for the target range in 1950. Only one measurement in each hemisphere is available in 1960, thus the target range is widened to mean ± 3 SD. After the 1950s, there is a discernible increasing trend in $\Delta^{14}C-CH_4$ observed in both hemispheres, which is attributed to biospheric $^{14}CH_4$ emissions after the 1950s and NPP $^{14}CH_4$ emissions after the 1970s (Lassey et al., 2007; Hmiel et al., 2020). Following 1980, the availability of $\Delta^{14}C-CH_4$ data increases, allowing for the estimation of global averages and uncertainties through spline fitting applied to data from both the northern and southern hemispheres. To account for measurement uncertainties reported in Hmiel et al. (2020), the spline fit is iteratively applied to the data spanning from the 1950s to the 2010s, with uncertainties randomly sampled 5000 times assuming a Gaussian distribution. Subsequently, the global average is determined as the mean of the northern and southern hemispheric data for each year. The uncertainty (1 SD) of the global averages is specified as one quarter of the inter-polar difference of spline fit for each year, assuming that the global averages fall within the two polar values with 95% CI. All the target ranges, constructed as above, encompass independent $\Delta^{14}C-CH_4$ measurements of firn air in Antarctica for 1940s–1970s and of air sampled globally for 1986–2000 in Lassey et al. (2007), except the data around 1976 ($\sim 160\%$), which exceeds our original upper bound at 1970 (117%). To cover a wider range considering the potential uncertainty of age of air both in Lassey et al. (2007) and Hmiel et al. (2020), we set 160% as our upper bound at 1970 (see Table S1).

When we convert global mean surface mole fractions to global atmospheric burden using the

conversion factor of 2.75 Tg CH₄ ppb⁻¹ (Prather et al. 2012), we did not account for the vertical gradient of $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, $\Delta^{14}\text{C-CH}_4$, as similar to previous isotope box model studies (e.g., Lassey et al. 2007; Schwietzke et al. 2016; Turner et al. 2017; Hmiel et al. 2020). Therefore, we assumed the same correction factor to account for reduced abundance in the stratosphere for ¹³CH₄, CH₃D, and ¹⁴CH₄ as for total CH₄ (“fill factor”; see Prather et al. (2012)), although the potential mean differences of $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, $\Delta^{14}\text{C-CH}_4$ between the stratosphere (~18 km altitudes in the mid latitudes) and surface are about +1.7‰ for $\delta^{13}\text{C-CH}_4$, +25‰ for $\delta\text{D-CH}_4$ and -55‰ for $\Delta^{14}\text{C-CH}_4$, respectively. Here, the stratospheric values for $\delta^{13}\text{C-CH}_4$ and $\delta\text{D-CH}_4$ were taken from Röckmann et al. (2011). No $\Delta^{14}\text{C-CH}_4$ data was available in the stratosphere, but since $\Delta^{14}\text{C-CH}_4$ cancels out isotopic fractionations due to loss processes, we can assume only the mean age of air in the stratosphere at 18 km (assumed to be 4 year; Waugh and Hall (2002)) would affect the mean difference from the tropospheric $\Delta^{14}\text{C-CH}_4$. Therefore, the potential maximum difference of $\Delta^{14}\text{C-CH}_4$ between stratosphere and surface over the analysis period is ~55‰ in the latter 1960s when the most rapid changes in $\Delta^{14}\text{C-CH}_4$ were observed (Fig. S1). Based on the above estimates and the mean air mass ratio between troposphere and stratosphere of 0.8 and 0.2 and corresponding CH₄ mole fractions of 1800 and 1550 ppb (equivalent to fill factor of 0.972; similar to Prather et al. (2012)’s estimate), the impact of potential biases using the same fill factor for ¹³CH₄, CH₃D, and ¹⁴CH₄ with CH₄ are estimated to be 0.3, 5, and 10‰ for the global mean $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, $\Delta^{14}\text{C-CH}_4$, respectively. These are comparable to the uncertainties of our prescribed atmospheric targets and are within the range of potential inter-laboratory differences in these measurements, where we find they do not significantly affect our posterior source fraction estimates (Fig. S12).

Prior emission scenarios

Table S2 shows our three a priori global CH₄ source scenarios using CEDS (version 2017-05-18) (Hoesly et al., 2018), EDGARv5.0 (Crippa et al., 2020) and EDGARv6.0 (Crippa et al., 2021), together with biomass burning (van Marle et al., 2017), natural biogenic (e.g., wetlands, freshwater) (Saunois et al. 2020) and geologic CH₄ sources (Etiope & Schwietzke, 2019; Hmiel et al., 2020). Our definition of total anthropogenic fossil emissions include “residential, commercial, and other sectors” (RCO) (Hoesly et al., 2018), whereas Saunois et al. (2020) separated these sectors into biomass burning sources. Thus our prior total anthropogenic fossil emissions are reported to be ~10 Tg/yr higher than those summarized in Saunois et al. (2020) (and vice versa in biomass burning emissions). We have investigated the sensitivity by moving the RCO emissions from prior anthropogenic FF emissions to prior BB emissions and confirmed that the posterior FF fractions are within the range of our base simulation results (Fig S5).

Comparison among different inventories

The fugitive CH₄ emission estimates in CEDS are based on EDGARv4.2 (Hoesly et al., 2018) and that EDGARv5.0 is based on EDGARv4.3.2 (Crippa et al., 2020), respectively, whose coal CH₄ emissions in China is reported to be overestimated (e.g., Maasakkers et al. (2019)). The total FF emissions in China in EDGARv6.0 for 2003–2012 is calculated to be 24.7 Tg CH₄ yr⁻¹, which is even 2.4 Tg CH₄ yr⁻¹ higher than those in EDGARv5.0. The difference is mostly explained by upward revision of coal emissions in China in EDGARv6.0, indicating that the overestimated China’s coal emission can still remain in EDGARv6.0. Similar overestimation in China is also seen when applied

the United Nations Framework Convention on Climate Change (UNFCCC)-based emission inventories (Scarpelli et al., 2020) to the atmospheric CH₄ inversion (Lu et al., 2021), although the global total anthropogenic emission estimates in UNFCCC are underestimated (IEA, 2022). The EDGARv7.0 is also available (https://edgar.jrc.ec.europa.eu/dataset_ghg70) (Crippa et al., 2022), but the values are almost the same as EDGARv6.0 until 2018 (Fig. 7). Also newer CEDS version 2021-04-21 is available (O'Rourke et al., 2021), which has higher fossil emissions than the version 2017-05-18 before 2000 but rather lowered after 2003 (Fig. 7). To investigate the robustness of our findings, further studies using other available prior emission inventories (Höglund-Isaksson et al., 2020; O'Rourke et al., 2021; IEA, 2022) are also required.

Spin-up procedure

Our spin-up procedure consists of two steps. First, to initiate the spin-up, the first target year, 1750, was initially compared to the atmospheric values assuming steady state. These atmospheric values were computed by the initial parameter ensembles generated from their full initial ranges using Latin hypercube sampling. The initial ensembles were then resampled based on the likelihood function in 1750. Subsequently, new uniform distributions were created for each parameter, based on the minimum and maximum values from the resampled distributions. These bounds were expanded by $\pm 2\%$ of the full initial ranges to encompass broader potential parameter solutions. Following this initial approximate optimization, a 50-yr spin-up simulation was further performed with the new parameter ensembles, during which all parameters were prescribed to be constant at their initial values. Finally, new ensembles were once again resampled based on the likelihood function in 1750. As a result of this two step procedure, the posterior ensembles are effectively determined while saving calculation time.

Reported posterior distributions

We report annual posterior CH₄ emissions and sinks and 20 parameter distributions for 1750–2015 for the CEDS, EDGARv5, and v6 scenarios individually (Fig. 1–6, Fig. S7, and Table S2) or averaged (Fig. 7–12, Table 2, Table 3, Fig. S2–S6, S8–S13, and Table S5). To calculate averages, we combined their posterior ensembles of each scenario (the each number of ensemble is 100,000) and calculate the statistics from the combined 300,000 ensembles.

Sensitivity tests

Here, we present the results of our sensitivity tests in Fig. S5–S6, S8–S12, which we did not fully discuss in the main text.

Only a few studies utilized atmospheric δD -CH₄ data to estimate global CH₄ budget as additional constraints (Rice et al., 2016; Fujita et al., 2020) and there is no recent studies that solely utilized δD -CH₄ with atmospheric CH₄, which restricted the detailed investigation on the impact of δD -CH₄. Experimental KIE^D_{OH} ranges 1.29 to 1.31 (Gierczak et al. 1997; Saueressig et al. 2001; Joelsson et al. 2016) except DeMore (1993) as 1.18. Based on the experimental KIE^D and sink fractions reported in Saunio et al. (2020), sink-weighted total KIE^D are calculated to be from 1.275 to 1.300, except when applying DeMore (1993) to KIE^D_{OH} as ~ 1.20 (Table S3). Considering the uncertainty in these estimated range and adopting the total KIE^D of 1.275 as our prior mean, used in Fujita et al. (2020), as

our prior mean, our prior base KIE^D range was set to be [1.25, 1.30]. Although this range does not fully cover the lower bound of total KIE^D based on KIE^D_{OH} of DeMore (1993), we attempted to cover the possibility of lower KIE^D scenario, less than 1.275. We explored the sensitivity by excluding the lower outlier of experimental KIE^D_{OH} (DeMore, 1993) and thus applying a narrower prior KIE^D range, [1.275, 1.300] (ParaRange #6, Table S3), the impact on the posterior sectorial source fraction being minimal (brown circle in Fig. S8).

Process models suggest stable global wetland emissions for 2000–2012 (Poulter et al., 2017), whereas recent ESMs indicate an increase up to ~ 30 Tg CH_4 yr^{-1} since pre-industrial (Paudel et al., 2016; Swart et al., 2019). Emissions from freshwater systems also vary widely (134–284 Tg CH_4 yr^{-1} in bottom-up and less than 30 Tg CH_4 yr^{-1} in top-down; Saunio et al. 2020). Our base scenario assumes time-invariant prior wetland and freshwater emissions from GCP bottom-up estimates (147 Tg CH_4 yr^{-1} and 159 Tg CH_4 yr^{-1} , respectively; Saunio et al. 2020), but with wider uncertainties including time-variable scaling factor (f_{natr_bio}). To explore the sensitivity of natural BIO (mainly wetlands and freshwater) emissions and trends, four alternative scenarios are also considered: (E) increasing wetland scenario from 147 Tg CH_4 yr^{-1} to 180 Tg CH_4 yr^{-1} for 1850–2015 (Paudel et al., 2016; Swart et al., 2019), (F) using VISIT (Ito, 2021) wetland emissions for 1901–2015 under Walter and Heimann (2000) emission scheme (G) same with (F) but for Cao et al. (1996) emission scheme, and (H) excluding freshwater emissions (IAWetBIOem and NoFRESHem #1). The results show minimal impact on recent CH_4 trends or mean source fractions for 2003–2012 (Fig S9). However, posterior anthropogenic BIO emissions in scenario E align more closely with CEDS prior emissions than in our base scenario that align with EDGARv5 and v6, despite total BIO emissions being consistent. Therefore, we have presented our posterior BIO emissions mainly for total BIO sources.

We found that “NoFRESHem” scenario (scenario H in Fig. S9) significantly impacts on the posterior source fractions when atmospheric $\Delta^{14}C$ - CH_4 data are not used as constraints, yielding higher FF fractions compared to those under the base prior emission setting (Fig. S11 and Table S5). This is primarily due to the lower posterior natural BIO and higher posterior GEO emissions under “ CH_4 only”, “ $CH_4 + \delta^{13}C$ ”, and “ $CH_4 + \delta D$ ” cases of no-freshwater scenario. For “Base” (all isotopic constraints) and “ $CH_4 + \Delta^{14}C$ ”, lower GEO emissions are well constrained by atmospheric $\Delta^{14}C$ - CH_4 data during historical period, regardless of the different prior natural BIO emissions, and thus the impact of no-freshwater scenario to the sectorial source fraction was minimal (Fig. S11 and Table S5).

The impact of potential inter-laboratory differences for isotopic measurements were also investigated (Fig. S12). When applying the upper bounds of $\delta^{13}C$ - CH_4 and δD - CH_4 inter-laboratory differences from our base setting (-0.31% from INSTAAR $\delta^{13}C$ - CH_4 scale and $+13.1\%$ from TU δD - CH_4 scale, presented in Fig. 2 of Umezawa et al. (2018)) to our observational target, our posterior FF fractions are derived to be 22.8 [18.9, 26.8] % and 25.4 [21.8, 28.7] %, respectively. These are within the uncertainty of our base results, 23.1 [19.0, 26.8] %, but careful attention should be paid to the impact of δD - CH_4 offset. No inter-laboratory comparison for $\Delta^{14}C$ - CH_4 measurements has been reported yet, but our sensitivity tests indicate our posterior estimate is robust within $\Delta^{14}C$ - CH_4 differences of $\pm 15\%$ (22.3 [18.0, 26.3] % and 24.2 [20.0, 28.1] % of posterior FF fraction, respectively). More careful consideration is required when assimilating historical raw $\delta^{13}C$ - CH_4 and δD - CH_4 data directly, whose inter-laboratory offsets are much larger than modern air samples (Umezawa et al. 2018). In this regard, we did not utilize respective raw $\delta^{13}C$ - CH_4 and δD - CH_4 data but applied wider multi-decadal target range for historical period, considering their potential large uncertainty to represent global averages.

Sink fractions derived from posterior total KIE^C and KIE^D

In the base scenario, we derive higher total KIE^C (1.0075 [1.0070, 1.0078]) and KIE^D (1.283 [1.274, 1.292]) for 2003–2012 than our prior mean setting (Fig. 3), which contributes to balancing the atmospheric ¹³CH₄ and CH₃D budgets. These parameters are also correlated with BIO and FF fractions in our base ensemble results, but not well correlated among the multiple sensitivity tests, compared with BIO $\delta^{13}\text{C}$ -CH₄ source signatures and NPP factors (ϕ). Here, we assume that the estimated total KIEs are equivalent to the sum of the experimental KIEs of the base setting (refer to Table S4) weighted by the respective sink fractions (similar approach in eq (5) of Röckmann et al., 2011). We also consider the atmospheric total CH₄ and stratospheric CH₄ lifetime to be 8.8 [8.3–9.4] years (our posterior mean for 2003–2012) and 152 ± 24 years (as adopted in Rigby et al. (2017)), respectively. Based on these assumptions, we derive the average sink fractions (and corresponding average atmospheric CH₄ lifetimes) as follows: $89 \pm 2\%$ (10.0 years) for tropospheric OH, $2.2 \pm 2.9\%$ (409 years) for soil oxidation, and $3.2 \pm 1.1\%$ (275 years) for tropospheric Cl. These estimates are generally consistent with bottom-up estimates in the Global Carbon Project (GCP) (Saunois et al., 2020). The poorly quantified sink contribution of tropospheric CH₄–Cl reactions is an important contributor to the uncertainty of CH₄ budget estimated from $\delta^{13}\text{C}$ -CH₄ due to its large KIE^C. Our estimated Cl contribution is lower than Allan et al. (2007), but higher than Gromov et al. (2018) where the tropospheric Cl contribution is almost zero. However, our derived sink-weighted KIEs are also sensitive to the selected KIEs in CH₄-OH reactions from the literature (Table S4). Further validation of Cl sink contributions and experimental studies on the KIEs are needed.

Comparison with Basu et al. (2022)

Basu et al. (2022) concluded from their sensitivity tests in the joint CH₄ and $\delta^{13}\text{C}$ -CH₄ inversion that the uncertainty in our knowledge of $\delta^{13}\text{C}$ -CH₄ source signatures is almost never a leading driver of uncertainty in sectorial source partitioning and larger uncertainties are related to tropospheric Cl and OH fractionation. However, their sensitivity test only evaluated the former impact by replacing prior $\delta^{13}\text{C}$ -CH₄ signature maps for selected three sources (biomass burning, ruminants, oil/natural gas, and wetlands), respectively, and they did not evaluate the whole combined uncertainty related to all $\delta^{13}\text{C}$ -CH₄ source signatures. It is also noted such $\delta^{13}\text{C}$ -CH₄ signature maps are generally produced by extrapolating and smoothing the limited direct measurements, whose potential uncertainty is large, particularly in the region where the data number is limited (e.g., tropics). Under the situation, we believe it is possible the uncertainty of $\delta^{13}\text{C}$ -CH₄ source signatures they have explored is still underestimated and more careful treatment is required for evaluating their impact comprehensively. Another reason of the contradiction between Basu et al. (2022) and ours could be due to different consideration of uncertainty for BIO $\delta^{13}\text{C}$ -CH₄ signatures. While our one-box model approach can only treat the global average explicitly, they evaluated the uncertainty of prior emission maps (i.e. wetland inundation extent) and source signature maps, separately. Therefore, if they combined all the uncertainties related to biogenic sources, the impact of uncertainty in their BIO isotopic source signature would be larger than they reported. Although our simple sensitivity tests suggest the impact of whole uncertainty of BIO $\delta^{13}\text{C}$ -CH₄ source signatures could be comparable to or even exceed that of total KIE^C when estimating the global CH₄ budget (Fig. 11), we acknowledge further investigation using a 3D model is more reliable, which can resolve their spatial information.

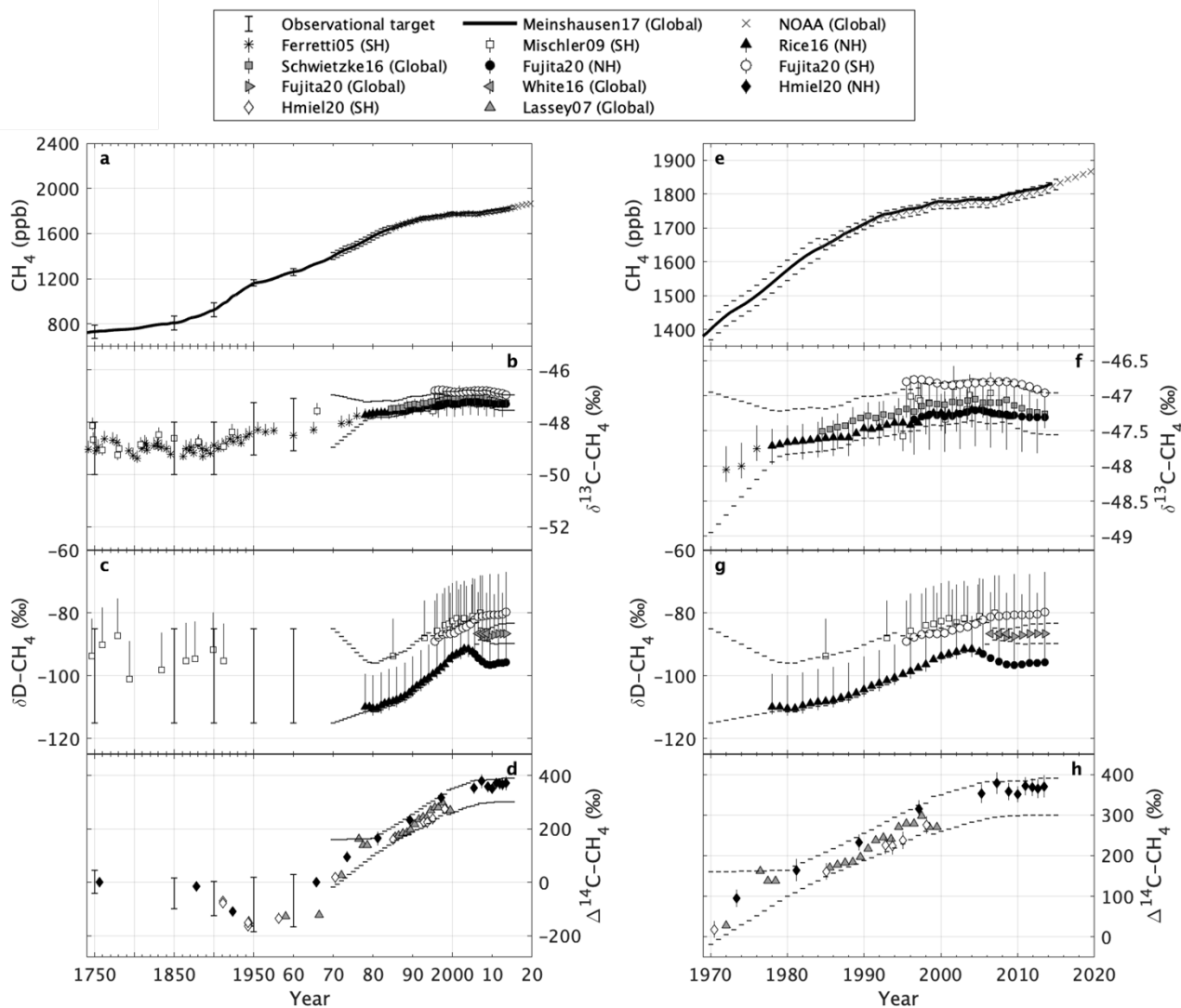


Fig. S1. Observed atmospheric CH_4 and isotopic histories compared to observation-based target ranges as shown in Fig. 1. Time series of observed atmospheric **a**, CH_4 , **b**, $\delta^{13}\text{C}-\text{CH}_4$, **c**, $\delta\text{D}-\text{CH}_4$, and **d**, $\Delta^{14}\text{C}-\text{CH}_4$ are used to define the target ranges. Note that the range represents minimum and maximum in Table S1 or ave. ± 3 SD when available. Enlarged axes for 1970–2020 are shown in **e–h**. Error bars in $\delta^{13}\text{C}-\text{CH}_4$ and $\delta\text{D}-\text{CH}_4$ represents inter-laboratory differences summarized in Umezawa et al. (2018). Note that the inter-laboratory differences of ice core and firn air samples whose CH_4 mole fraction is low could be larger than those of modern surface air samples (Umezawa et al., 2018); thus wider target ranges are roughly specified for the period of 1750–1970 (Text S1). Error bars in $\Delta^{14}\text{C}-\text{CH}_4$ represents 3 SD of each measurement in Hmiel et al. (2020).

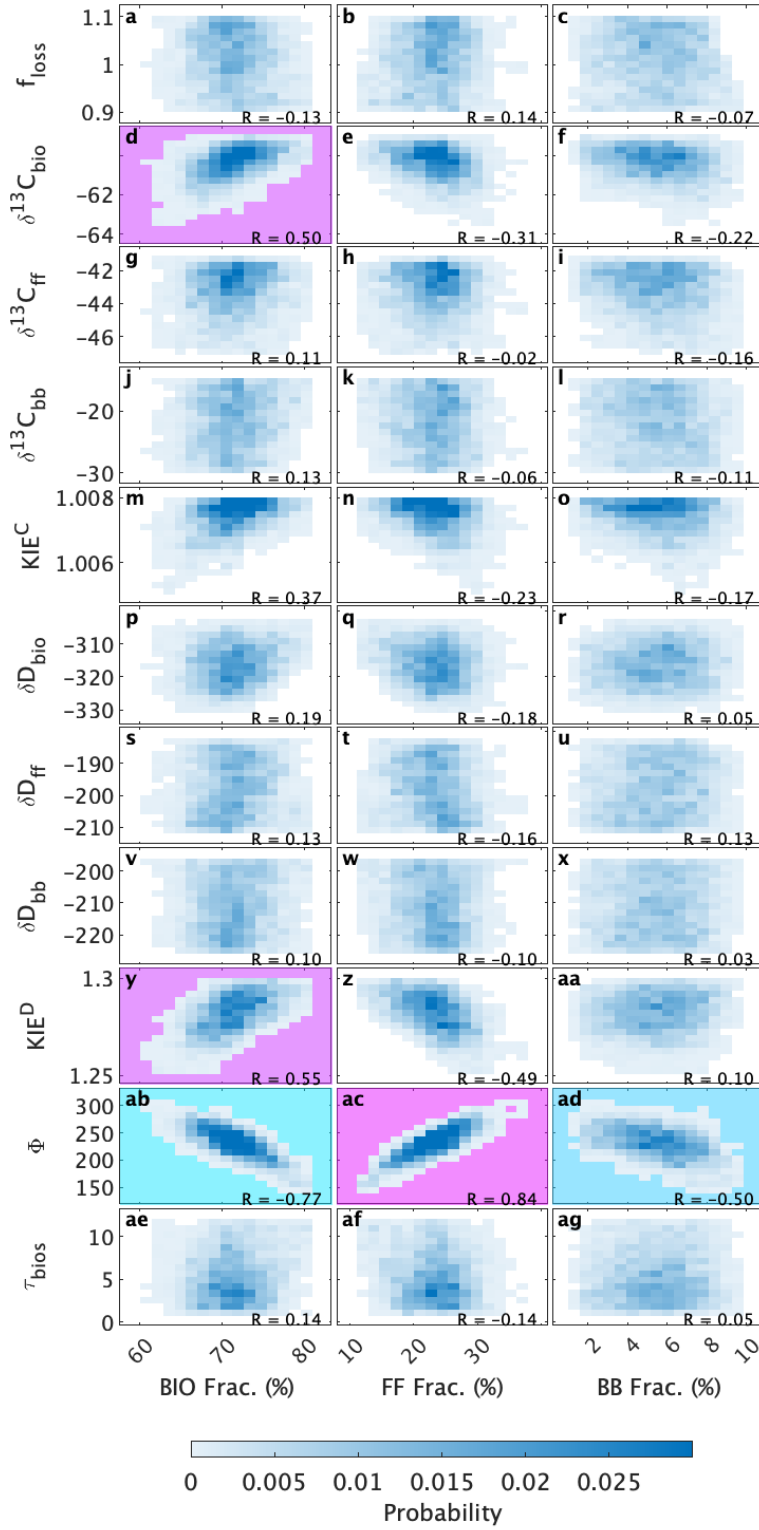


Fig. S2. Same as Fig. 8 in the main text, but for global total CH_4 emissions (tot), biogenic emissions (bio), fossil fuel emissions (ff), and biomass burning emissions (bb) vs respective parameters.

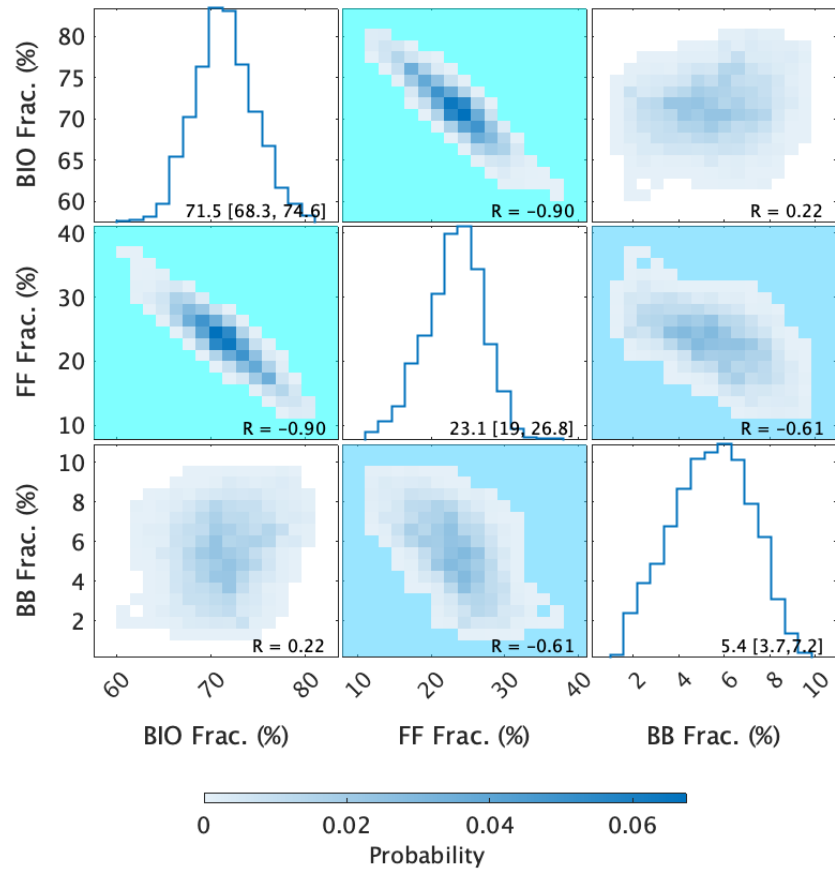


Fig. S3. Same as Fig. 8, but for among sectorial source fractions. The numbers on each histogram show mean and 68% CI for each posterior source fractions.

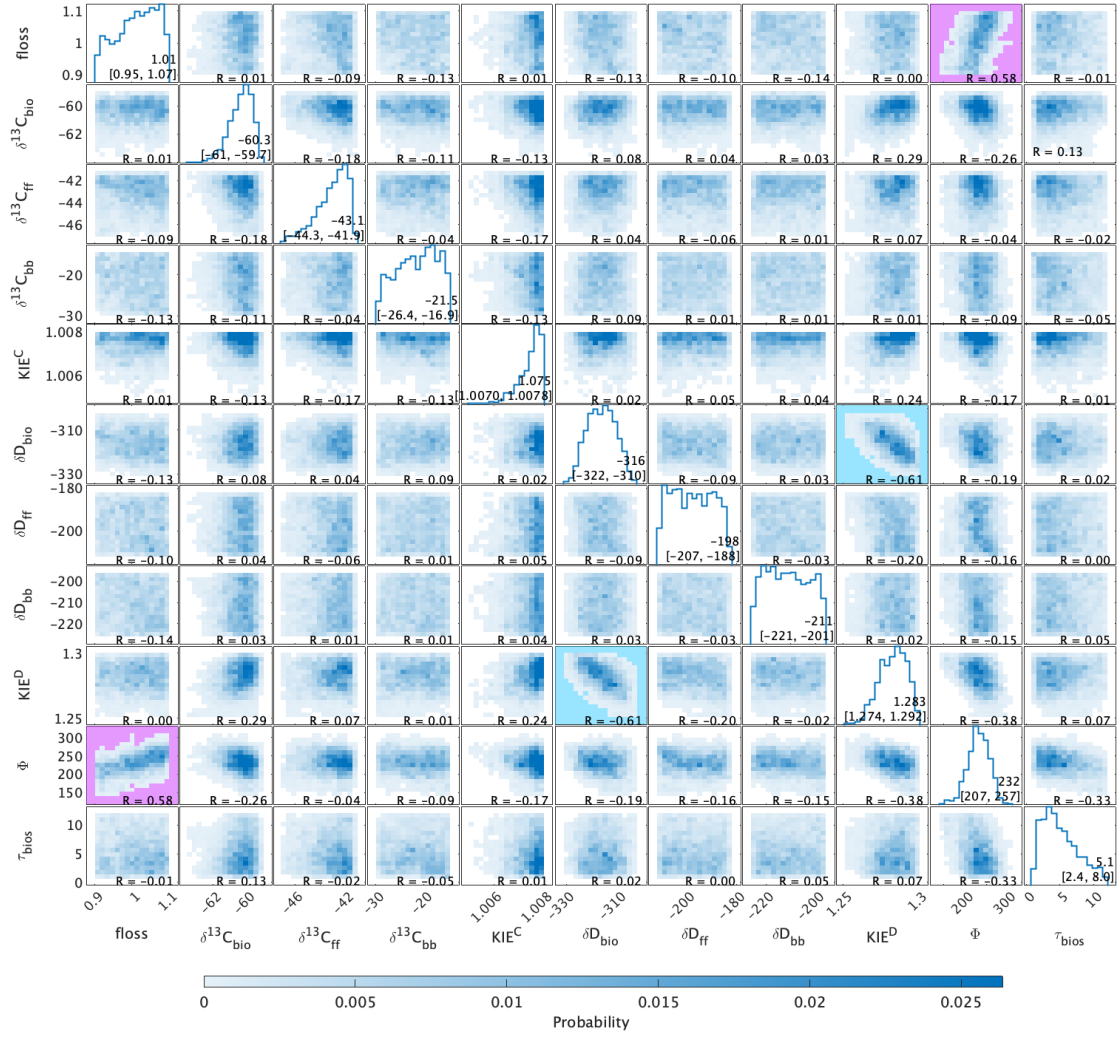


Fig. S4. Same as Fig. 8, but for among respective parameters. The numbers on each histogram show mean and 68% CI for each posterior parameter.

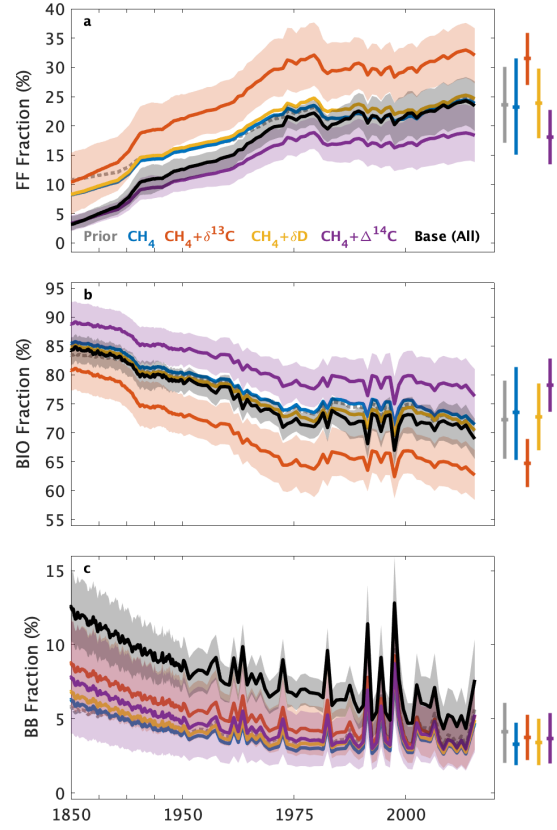


Fig. S5. Timeseries of posterior global sectorial source fractions for total fossil fuel (FF) **(a)**, total biogenic (BIO) **(b)**, and biomass burning **(c)** sources. Different colors indicate different observational constraints scenarios, as presented in Fig. 10. Posterior mean values are shown in solid lines and colored bands (only for “CH₄+δ¹³C”, “CH₄+Δ¹⁴C”, and base results) mark the 68% CI. Colored bars in right show average and 68% CI for 2003–2012. Dotted gray lines (mostly overlapped by CH₄-only results) and gray bars show prior estimates.

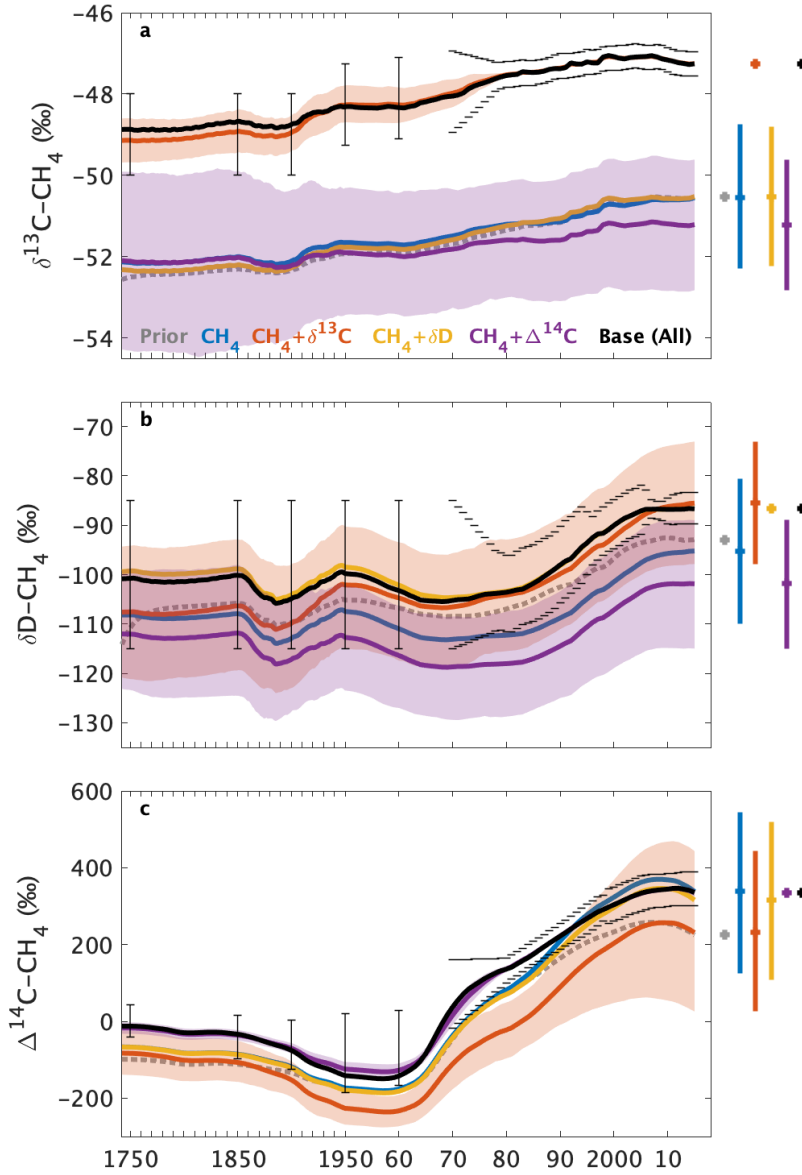


Fig. S6. Same as Fig. S5 in the main text, but for simulated atmospheric isotopic histories compared to observation-based target ranges (same as Fig. 1). Color bands (68% CI) are presented only for “CH₄+δ¹³C” and “CH₄+Δ¹⁴C”. Here simulation results of atmospheric CH₄ is not presented since the difference of simulate CH₄ among the presented scenarios is negligible, except Prior scenario (see Fig. 1). Due to the significantly higher atmospheric CH₄ mole fraction in the Prior scenario (~1.3 times higher than the other scenarios), the influence of rapid NPP ¹⁴CH₄ emissions after the 1980s on atmospheric Δ¹⁴C-CH₄ variations is diluted in the Prior scenario, in contrast to the other scenarios. Consequently, despite similar fossil source fractions and NPP ¹⁴CH₄ emissions with “CH₄-only” scenario after 1980s (see Fig. S5), the rate of increase in atmospheric Δ¹⁴C-CH₄ shows smaller in the Prior scenario than “CH₄-only” scenario.

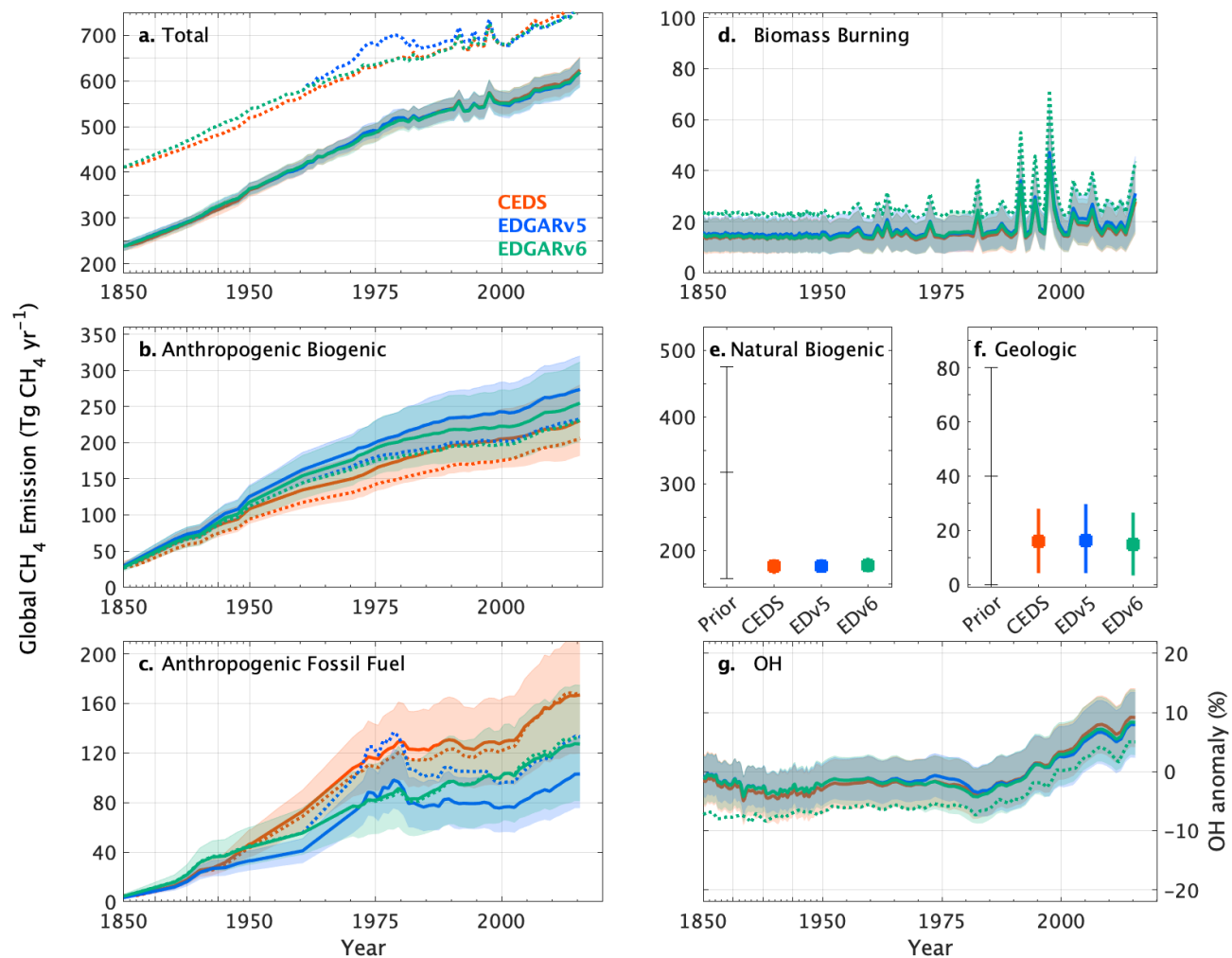


Fig. S7. Same as Fig. 6 in the main text, but for “CH₄-only” case.

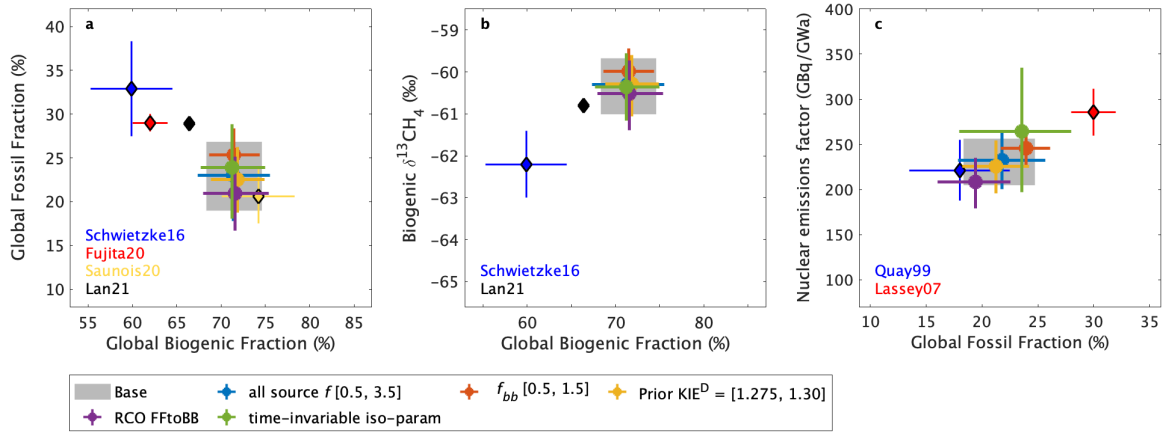


Fig. S8. Same as Fig. 10 in the main text, but for sensitivity tests on the different prior ranges of source scaling factors and KIE^D, different prior BB and FF source allocations (by allocating “RCO” sources from prior FF sources to prior BB sources), and different time-variations of prior isotopic parameters (i.e. α of all isotopic source signatures, KIEs, τ_{bios} , and ϕ are fixed at 0.3, same as geologic parameters).

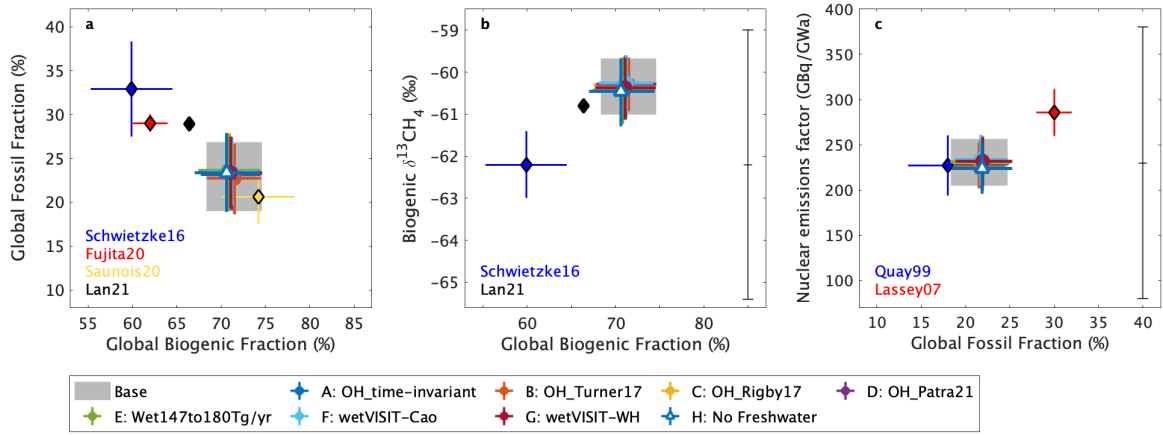


Fig. S9. Same as Fig. 10, but for sensitivity tests on the difference of global OH trends between global chemistry-climate models and methyl chloroform (MCF)-based estimates and difference of prior natural BIO emission settings. All sensitivity test results are barely changed from the base simulation in terms of mean source fractions, which are shown to be overwrapped here.

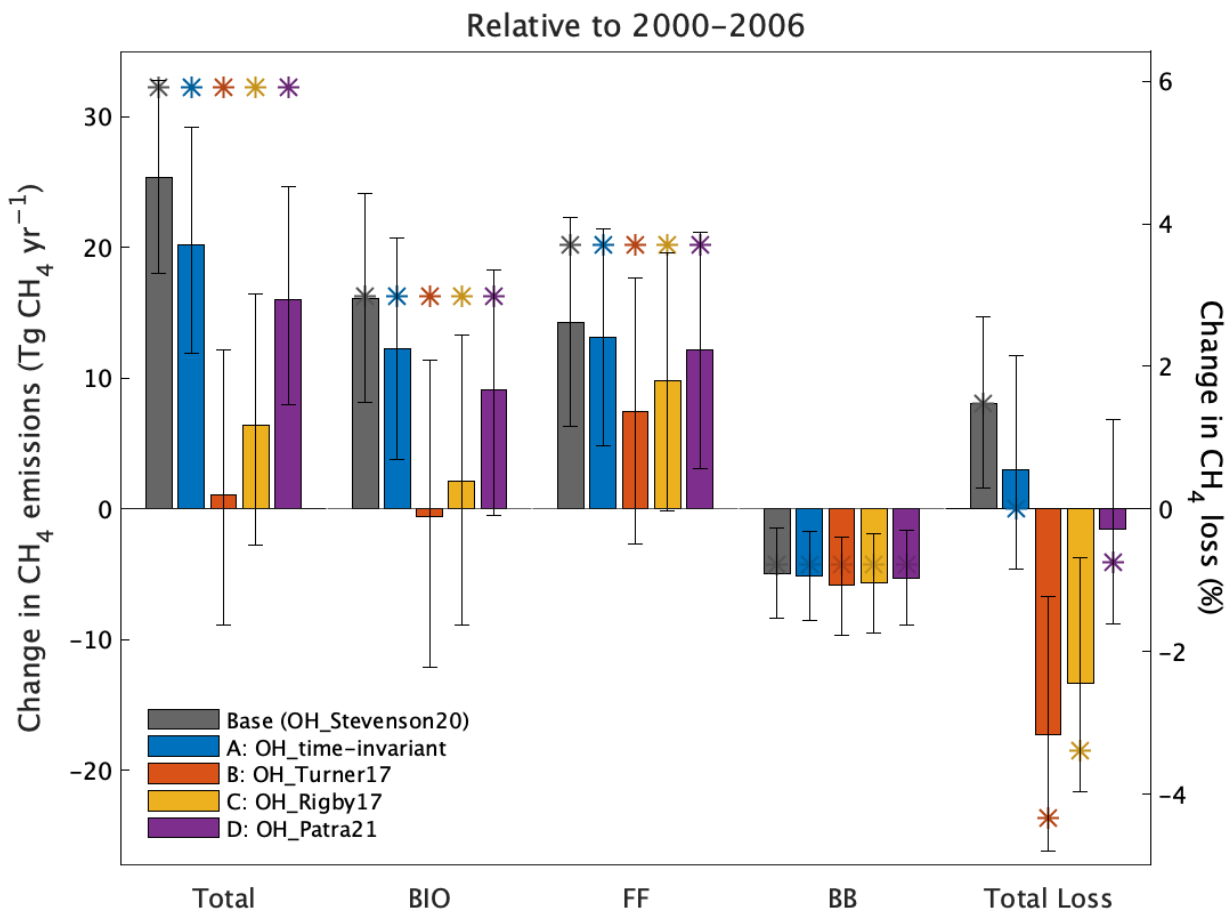


Fig. S10. Changes of CH_4 emissions and loss rates averaged between 2000–2006 and 2007–2013. Base result represents our posterior base scenario and A–D (same as Fig. S9) represent sensitivity test results examining the influence of different prior OH scenarios. The impact of different prior natural BIO emission scenarios (scenario E–H) on the posterior emission and loss changes for the period are minimal; thus not shown here. Error bars represent 68% CI. Prior mean values are represented by colored asterisks. The vertical range of change in total CH_4 loss rate (right axis) is adjusted corresponding to the percentage of change in total CH_4 emissions.

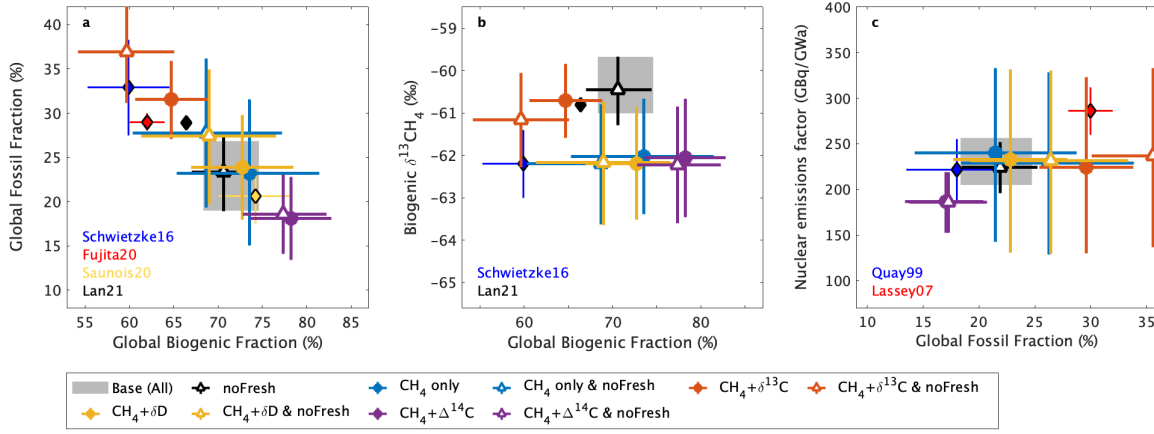


Fig. S11. Same as Fig. 10, but for sensitivity tests on the “NoFRESHem” scenario under different isotopic constraints. “noFresh” (white triangle with black edge) represents the same as “H: No freshwater” in Fig S9. “ CH_4 only”, “ $\text{CH}_4 + \delta^{13}\text{C}$ ”, “ $\text{CH}_4 + \delta\text{D}$ ”, and “ $\text{CH}_4 + \Delta^{14}\text{C}$ ” represent the same as those in Fig. 10.

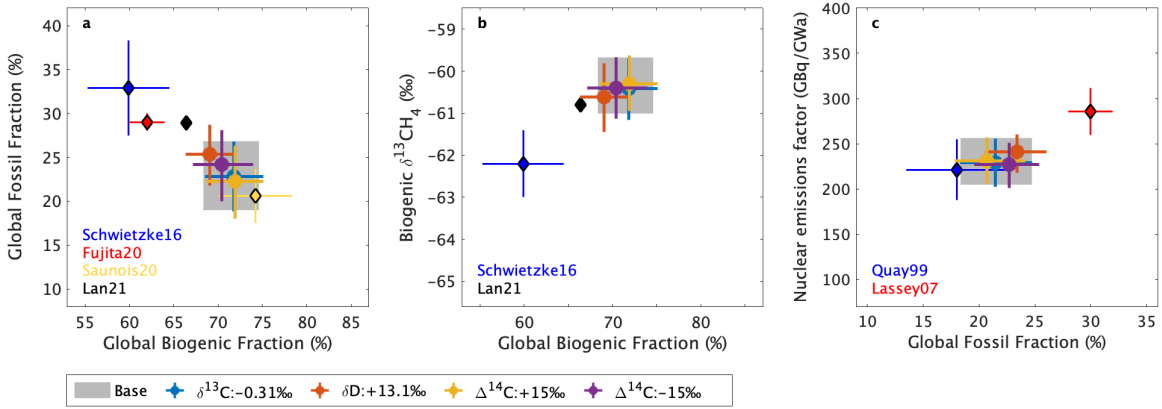


Fig. S12. Same as Fig. 10, but for sensitivity tests that applied the potential offset to the observational target for examining the influence of inter-laboratory differences in $\delta^{13}\text{C}-\text{CH}_4$, $\delta\text{D}-\text{CH}_4$, and $\Delta^{14}\text{C}-\text{CH}_4$ measurements.

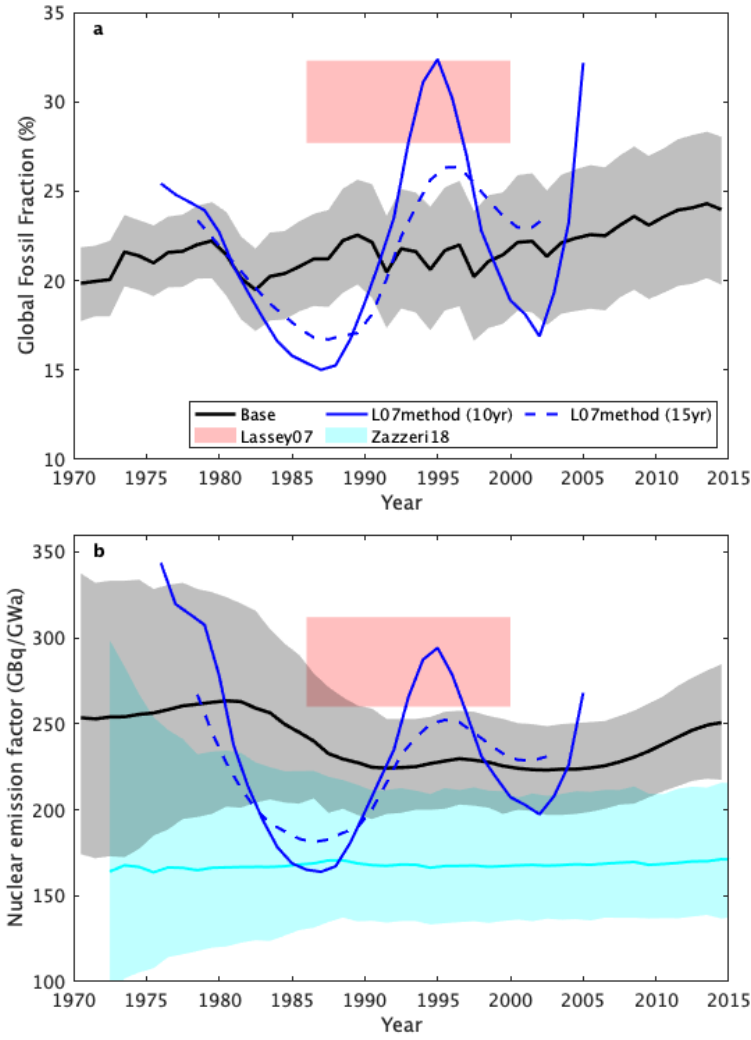


Fig. S13. Timeseries of global fossil source fractions (a), and timeseries of global nuclear power plant $^{14}\text{CH}_4$ emission factor (b). Our posterior mean and 68% CI under the base simulation are represented by black lines and gray shades. Results from “L07method” (see section 4 in the main text) using our model dataset with 10-yr windows and 15-yr windows are represented by blue solid and blue dash lines, respectively. Results from Lassey et al. (2007) for 1986–2000 is represented by red square (mean ± 1 SD). Our updated observational estimates updated from Zazzeri et al. (2018) are represented by light cyan line and shades (median and 95% CI).

Table S1. Observational target years and ranges for atmospheric CH₄, $\delta^{13}\text{C-CH}_4$, $\delta\text{D-CH}_4$, and $\Delta^{14}\text{C-CH}_4$.

Target year	CH ₄ (ppb)		$\delta^{13}\text{C-CH}_4$ (‰)				$\delta\text{D-CH}_4$ (‰)				$\Delta^{14}\text{C-CH}_4$ (‰)			
	ave.	1 s.d.	ave.	1 s.d.	min.	max.	ave.	1 s.d.	min.	max.	ave.	1 s.d.	min.	max.
1750.0	731.2	20	-	-	-50.00	-48.00	-	-	-115.0	-85.0	-	-	-40.9	43.9
1850.0	808.2	20	-	-	-50.00	-48.00	-	-	-115.0	-85.0	-	-	-97.2	16.5
1900.0	923.9	20	-	-	-50.00	-48.00	-	-	-115.0	-85.0	-	-	-125.4	2.8
1950.0	1162.1	10	-	-	-49.25	-47.25	-	-	-115.0	-85.0	-	-	-185.8	19.6
1960.0	1260.7	10	-	-	-49.10	-47.10	-	-	-115.0	-85.0	-	-	-167.3	29.0
1970.0	1400.2	10	-	-	-48.95	-46.95	-	-	-115.0	-85.0	-	-	-18.0	160.0
1971.0	1421.1	10	-	-	-48.82	-46.98	-	-	-114.6	-86.3	-	-	-6.2	160.4
1972.0	1440.2	10	-	-	-48.70	-47.02	-	-	-114.1	-87.5	-	-	5.5	160.9
1973.0	1456.1	10	-	-	-48.57	-47.05	-	-	-113.7	-88.8	-	-	17.3	161.3
1974.0	1469.5	10	-	-	-48.44	-47.08	-	-	-113.2	-90.0	-	-	29.1	161.7
1975.0	1483.9	10	-	-	-48.32	-47.11	-	-	-112.8	-91.3	-	-	40.9	162.2
1976.0	1500.4	10	-	-	-48.19	-47.15	-	-	-112.3	-92.5	-	-	52.6	162.6
1977.0	1518.4	10	-	-	-48.07	-47.18	-	-	-111.9	-93.8	-	-	64.4	163.0
1978.0	1537.3	10	-	-	-47.94	-47.21	-	-	-111.5	-95.0	-	-	76.2	163.4
1979.0	1556.5	10	-47.54	0.11	-	-	-103.4	2.6	-	-	-	-	87.9	163.9
1980.0	1575.6	10	-47.52	0.11	-	-	-103.8	2.6	-	-	132.0	10.8	-	-
1981.0	1593.8	10	-47.51	0.11	-	-	-103.8	2.6	-	-	141.4	10.7	-	-
1982.0	1610.7	10	-47.50	0.11	-	-	-103.1	2.6	-	-	150.7	10.5	-	-
1983.0	1625.7	10	-47.49	0.11	-	-	-102.4	2.6	-	-	160.0	10.3	-	-
1984.0	1638.1	10	-47.49	0.10	-	-	-102.0	2.6	-	-	169.2	10.1	-	-
1985.0	1650.9	5	-47.49	0.10	-	-	-101.7	2.6	-	-	178.4	9.9	-	-
1986.0	1663.7	5	-47.46	0.10	-	-	-101.1	2.6	-	-	187.0	10.1	-	-
1987.0	1676.5	5	-47.44	0.10	-	-	-100.5	2.6	-	-	195.4	10.3	-	-
1988.0	1687.9	5	-47.39	0.10	-	-	-99.9	2.6	-	-	203.8	10.5	-	-
1989.0	1698.8	5	-47.34	0.10	-	-	-99.1	2.6	-	-	212.3	10.7	-	-

1990.0	1709.5	5	-47.31	0.10	-	-	-98.0	2.6	-	-	221.0	11.0	-	-
1991.0	1719.7	5	-47.31	0.10	-	-	-97.0	2.6	-	-	229.9	11.5	-	-
1992.0	1730.2	5	-47.30	0.10	-	-	-96.0	2.6	-	-	238.8	11.9	-	-
1993.0	1736.1	5	-47.28	0.10	-	-	-95.1	2.6	-	-	247.8	12.3	-	-
1994.0	1739.4	5	-47.26	0.10	-	-	-94.1	2.6	-	-	257.0	12.5	-	-
1995.0	1745.6	5	-47.21	0.10	-	-	-93.6	2.1	-	-	266.5	12.7	-	-
1996.0	1750.2	5	-47.21	0.10	-	-	-93.2	1.6	-	-	276.5	12.5	-	-
1997.0	1753.0	5	-47.19	0.10	-	-	-92.1	1.6	-	-	286.5	12.4	-	-
1998.0	1760.2	5	-47.13	0.10	-	-	-91.2	1.6	-	-	294.5	11.4	-	-
1999.0	1769.1	5	-47.12	0.10	-	-	-90.4	1.6	-	-	299.9	15.3	-	-
2000.0	1772.8	5	-47.13	0.10	-	-	-89.8	1.6	-	-	305.0	14.8	-	-
2001.0	1772.1	5	-47.11	0.10	-	-	-89.2	1.6	-	-	310.7	15.0	-	-
2002.0	1772.0	5	-47.09	0.10	-	-	-88.5	1.6	-	-	316.3	15.0	-	-
2003.0	1775.1	5	-47.08	0.10	-	-	-87.8	1.6	-	-	321.8	14.8	-	-
2004.0	1777.3	5	-47.06	0.10	-	-	-87.5	1.6	-	-	327.2	15.0	-	-
2005.0	1775.6	5	-47.07	0.10	-	-	-86.9	1.6	-	-	332.5	15.4	-	-
2006.0	1774.4	5	-47.10	0.10	-	-	-86.4	1.2	-	-	336.2	15.1	-	-
2007.0	1778.0	5	-47.10	0.10	-	-	-86.7	0.7	-	-	338.4	14.6	-	-
2008.0	1784.2	5	-47.08	0.10	-	-	-86.9	0.6	-	-	339.8	14.3	-	-
2009.0	1790.3	5	-47.10	0.10	-	-	-87.1	0.7	-	-	340.6	14.3	-	-
2010.0	1796.2	5	-47.16	0.10	-	-	-87.1	0.9	-	-	341.5	14.3	-	-
2011.0	1801.0	5	-47.21	0.10	-	-	-86.9	1.0	-	-	342.4	14.1	-	-
2012.0	1805.6	5	-47.24	0.10	-	-	-86.6	1.0	-	-	343.5	14.5	-	-
2013.0	1810.6	5	-47.25	0.10	-	-	-86.5	1.1	-	-	345.1	15.0	-	-
2014.0	1817.9	5	-47.26	0.10	-	-	-86.5	1.1	-	-	345.1	15.0	-	-
2015.0	1828.3	5	-47.26	0.10	-	-	-86.5	1.1	-	-	345.1	15.0	-	-

Note: Our atmospheric $\delta^{13}\text{C-CH}_4$ target for 1984–2013 is taken from Fig S3 of Schwietzke et al. (2016). The target in 2014.0 and 2015.0 is repeated by the annual average in 2013. We confirmed the consistency with updated global annual $\delta^{13}\text{C-CH}_4$ averages for 1999–2013 by INSTAAR (Michel et al., 2021), the difference being within $\pm 0.01\text{‰}$. We also performed the sensitivity test by replacing the $\delta^{13}\text{C-CH}_4$ target for 1999–2015 by the updated dataset (Michel et al. 2021) and confirmed the impact on the posterior results is minimal.

Table S2. A priori global CH₄ emissions for CEDS, EDGARv5, and EDGARv6 scenarios for selected periods. The bold character represents main source categories and italic character represents sub-categories. The units are teragrams of methane per year.

Source Category	1900–1950			1986–2000			2003–2012		
	CEDS	EDGA Rv5	EDGA Rv6	CEDS	EDGA Rv5	EDGA Rv6	CEDS	EDGA Rv5	EDGA Rv6
Anthropogenic biogenic	74	86	86	170	200	195	191	218	214
<i>Agriculture</i>	<i>57</i>	<i>75</i>	<i>75</i>	<i>119</i>	<i>138</i>	<i>139</i>	<i>129</i>	<i>146</i>	<i>147</i>
<i>Waste</i>	<i>17</i>	<i>11</i>	<i>11</i>	<i>51</i>	<i>62</i>	<i>56</i>	<i>62</i>	<i>72</i>	<i>68</i>
Natural biogenic^a	317								
<i>Wetlands</i>	<i>147</i>								
<i>Freshwater</i>	<i>159</i>								
<i>Termites</i>	<i>9</i>								
<i>Wild animals</i>	<i>2</i>								
Anthropogenic fossil	29	36	36	120	103	97	152	115	120
<i>Energy (oil, gas, & coal)</i>	<i>24</i>	<i>28</i>	<i>28</i>	<i>107</i>	<i>89</i>	<i>82</i>	<i>139</i>	<i>100</i>	<i>106</i>
<i>Residential + others</i>	<i>6</i>	<i>9</i>	<i>9</i>	<i>14</i>	<i>15</i>	<i>15</i>	<i>13</i>	<i>14</i>	<i>14</i>
Geologic^b	40								
Biomass burning^c	23			32			29		
Total biogenic	391	403	403	487	517	512	508	535	531
Total fossil fuel	69	76	76	160	143	137	192	155	160
Total (Total biogenic + Total fossil fuel + Biomass burning)	484	502	502	680	693	681	729	719	720

^a Table 3 of Saunio et al. (2020) bottom-up estimates for the period 2000–2009.

^b Mean of our prior ranges (see Table 1 in the main text).

^c van Marle et al. (2017). Here it is multiplied by 2.0 (i.e., mean of our prior ranges of f_{bb} , see Table 1).

Table S3. Sensitivity tests of our one-box model inversions.

Scenario	Focus	Description	Fig.	Legend name
Base		The base inversion using all observational constraints and default parameter settings.	-	Base (All)
Obs	Observational targets	Seven inversions under different combinations of observational isotopic constraints listed below.		
	1	CH ₄ only	Fig. 10	CH ₄ only
	2	CH ₄ and $\delta^{13}\text{C}$ -CH ₄	Fig. 10	CH ₄ + $\delta^{13}\text{C}$
	3	CH ₄ and δD -CH ₄	Fig. 10	CH ₄ + δD
	4	CH ₄ and $\Delta^{14}\text{C}$ -CH ₄	Fig. 10	CH ₄ + $\Delta^{14}\text{C}$
	5	CH ₄ , $\delta^{13}\text{C}$ -CH ₄ , and δD -CH ₄	Fig. 10	CH ₄ + $\delta^{13}\text{C}$ + δD
	6	CH ₄ , $\delta^{13}\text{C}$ -CH ₄ , and $\Delta^{14}\text{C}$ -CH ₄	Fig. 10	CH ₄ + $\delta^{13}\text{C}$ + $\Delta^{14}\text{C}$
	7	CH ₄ , δD -CH ₄ , and $\Delta^{14}\text{C}$ -CH ₄	Fig. 10	CH ₄ + δD + $\Delta^{14}\text{C}$
ObsIsoILD	Observational targets	Four inversions under potential maximum inter-laboratory differences of $\delta^{13}\text{C}$ -CH ₄ , δD -CH ₄ , and $\Delta^{14}\text{C}$ -CH ₄ measurements from our base targets listed below.		
	1	$\delta^{13}\text{C}$ -CH ₄ Adding offset -0.31‰ to the observational target (from INSTAAR to UB; see Fig 2a of Umezawa et al. 2018).	Fig. S12	$\delta^{13}\text{C}$: -0.31‰
	2	δD -CH ₄ Adding offset +13.1‰ to the observational target (from TU to IMAU; see Fig 2b of Umezawa et al. 2018).	Fig. S12	δD : +13.1‰
	3	$\Delta^{14}\text{C}$ -CH ₄ Adding offset +15‰ to the observational target.	Fig. S12	$\Delta^{14}\text{C}$: +15‰
	4	$\Delta^{14}\text{C}$ -CH ₄ Adding offset -15‰ to the observational target.	Fig. S12	$\Delta^{14}\text{C}$: -15‰
ParaRange	Parameters	15 inversions under different prior parameter ranges listed as below.		
	1	<i>BIO</i> $\delta^{13}\text{C}$ -CH ₄ ^a Fix at -62.2	Fig. 11	Fixed prior <i>BIO</i> $\delta^{13}\text{CH}_4$
	2	<i>BIO</i> $\delta^{13}\text{C}$ -CH ₄ & <i>KIE</i> ^C Fix at -62.2 and 1.0065	Fig. 11	Fixed prior <i>BIO</i> $\delta^{13}\text{CH}_4$ & <i>KIE</i> ^C
	3	<i>BIO</i> $\delta^{13}\text{C}$ -CH ₄ & <i>KIE</i> ^C Fix at -62.2 and 1.0065 under “CH ₄ + $\delta^{13}\text{C}$ ” case	Fig. 11	CH ₄ + $\delta^{13}\text{C}$ +Fixed prior <i>BIO</i> $\delta^{13}\text{CH}_4$ & <i>KIE</i> ^C
	4	<i>KIE</i> ^C Fixed at 1.0065	Fig. 11	Fixed prior <i>KIE</i> ^C
	5	<i>KIE</i> ^C Wider prior range [1.005, 1.010]	Fig. 11	Prior <i>KIE</i> ^C = [1.005, 1.010]
	6	<i>KIE</i> ^D Fixed at 1.275	Fig. 11	Fixed prior <i>KIE</i> ^D
	7	<i>KIE</i> ^D Narrower prior range [1.275, 1.300]	Fig. S8	Prior <i>KIE</i> ^D = [1.275, 1.30]
	8	ϕ Different mean with same prior range 286 [136, 436]	Fig. 12	Prior ϕ = [136, 436]
	9	ϕ Different mean with wider prior range 300 [0, 600]	Fig. 12	Prior ϕ = [0, 600]
	10	ϕ Fixed at 286	Fig. 12	Prior ϕ = 286 fixed
	11	ϕ Different mean with wider prior range 300 [0, 600] under “CH ₄ + $\Delta^{14}\text{C}$ ” case	Fig. 12	CH ₄ + $\Delta^{14}\text{C}$ & Prior ϕ = [0, 600]
	12	ϕ Fixed at 286 under “CH ₄ + $\Delta^{14}\text{C}$ ” case	Fig. 12	CH ₄ + $\Delta^{14}\text{C}$ & Prior ϕ = 286 fixed
	13	τ_{bias} Fixed at 6.5 yr	Fig. 12	Fixed prior τ_{bias}

Table S3. (continued)

14	f_{bb}	Narrower prior range [0.5, 1.5]	Fig. S8	f_{bb} [0.5, 1.5]
15	f_{anth_bio} , f_{natr_bio} & f_{anth_ff}	Wider prior range [0.5, 3.5]	Fig. S8	all source f [0.5, 3.5]
IsoParaFixT	All isotopic source signatures, $KIEs$, τ_{bios} , & ϕ	Time variation parameter α is fixed at 0.3	Fig. S8	time-invariable iso-param
RCOem	Prior BB and anthropogenic FF emissions	Allocating “RCO” emissions from prior FF to prior BB emissions. ^b	Fig. S8	RCO FFtoBB
OHTrend	Prior sinks	Four inversions under different prior OH anomaly scenarios listed below.		
1		Time-invariant OH for 1750–2015	Fig. S9 & S10	A: OH_time_invariant
2		MCF-based OH from Turner et al. (2017) for 1980–2015	Fig. S9 & S10	B: OH_Turner17
3		MCF-based OH from Rigby et al. (2017) for 1980–2015	Fig. S9 & S10	C: OH_Rigby17
4		MCF-based OH from Patra et al. (2021) for 1995–2015	Fig. S9 & S10	D: OH_Patra21
IAWetBIOem	Prior natural BIO emissions	Three inversions under different prior interannual wetland emission scenarios listed as below.		
1		Increasing wetland emissions from 147 TgCH ₄ yr ⁻¹ to 180 TgCH ₄ yr ⁻¹ for 1850–2015 (Paudel et al., 2016; Swart et al., 2019)	Fig. S9	E: Wet147to180Tg/yr
2		VISIT (Ito, 2021) wetland emissions for 1901–2015 under Walter and Heimann (2000) emission scheme	Fig. S9	F: wetVISIT-Cao
3		VISIT (Ito, 2021) wetland emissions for 1901–2015 under Cao et al. (1996) emission scheme	Fig. S9	G: wetVISIT-WH
NoFRESHem	Prior natural BIO emissions	Five inversions with freshwater emissions (159 Tg/yr) excluded under different observational isotopic constraints listed below.		
1		All isotopic constraints	Fig. S9 & S11	H: No Freshwater (Fig. S9) or noFresh (Fig. S11)
2		CH ₄ only	Fig. S11 & Table S5	CH ₄ only & noFresh
3		CH ₄ and $\delta^{13}C$ -CH ₄	Fig. S11 & Table S5	CH ₄ + $\delta^{13}C$ & noFresh
4		CH ₄ and δD -CH ₄	Fig. S11 & Table S5	CH ₄ + δD & noFresh
5		CH ₄ and $\Delta^{14}C$ -CH ₄	Fig. S11 & Table S5	CH ₄ + $\Delta^{14}C$ & noFresh

^a Both $\delta^{13}C_{anth_bio}$ and $\delta^{13}C_{natr_bio}$

^b To find plausible solutions of posterior BB emissions reported among previous studies, we have also changed the range of f_{bb} from [0.5, 3.5] to [0.5, 2.0].

Table S4. Summary of experimental KIEs based on the literatures.

Scenario	KIE ^C							KIE ^D						
	OH	soil	Cl	strat.	total			OH	soil	Cl	strat.	total		
					S20: min trop. Cl	S20 ave.	S20: max trop. Cl					S20: min trop. Cl	S20 ave.	S20: max trop. Cl
Base	1.00465 ^a	1.0201 ^b	1.0666 ^c	1.0147 ^d	1.0060	1.0070	1.0092	1.292 ⁱ	1.083 ^j	1.527 ^k	1.137 ^d	1.278	1.279	1.274
KIEC_OH_S01	1.0039 ^e	*	*	*	1.0054	1.0064	1.0086	*	*	*	*	*	*	*
KIEC_OH_C90	1.0054 ^f	*	*	*	1.0067	1.0077	1.0099	*	*	*	*	*	*	*
KIEC_Cl_T00	*	*	1.0625 ^g	*	1.0060	1.0069	1.0090	*	*	*	*	*	*	*
KIEC_strat_R03	*	*	*	1.0154 ^h	1.0060	1.0070	1.0092	*	*	*	*	*	*	*
KIED_OH_S01	*	*	*	*	*	*	*	1.294 ^f	*	*	*	1.276	1.280	1.289
KIED_OH_J16	*	*	*	*	*	*	*	1.31 ^l	*	*	*	1.290	1.294	1.303
KIED_OH_D93	*	*	*	*	*	*	*	1.18 ^m	*	*	*	1.174	1.179	1.192
KIED_Cl_T00	*	*	*	*	*	*	*	*	*	1.479 ^g	*	1.274	1.278	1.285
KIED_Cl_R03	*	*	*	*	*	*	*	*	*	*	1.153 ^h	1.280	1.279	1.289

^a Average of Cantrell et al. (1990) and Saueressig et al. (2001); ^b Average of Snover and Quay (2000), Tyler et al. (1994), and Reeburgh et al. (1997); ^c Saueressig et al. (1995); ^d Röckmann et al. (2011); ^e Saueressig et al. (2001); ^f Cantrell et al. (1990); ^g Tyler et al. (2000); ^h Rice et al. (2003); ⁱ Gierczak et al. (1997); ^j Snover and Quay (2000); ^k Saueressig et al. (1996); ^l Joelsson et al. (2016); ^m DeMore (1993)

Note: The total KIEs are calculated using bottom-up estimates of respective sink fractions reported in Saunio et al. (2020) (S20) for 2000–2009 and presented respective KIEs of tropospheric OH reaction, soil oxidation, tropospheric Cl reaction, and stratospheric destruction processes in the literatures (Cantrell et al., 1990; DeMore, 1993; Tyler et al., 1994; Saueressig et al., 1995, 1996; Gierczak et al., 1997; Reeburgh et al., 1997; Snover & Quay, 2000; Tyler et al., 2000; Saueressig et al., 2001; Rice et al. 2003; Röckmann et al., 2011; Joelsson et al., 2016). For the sink fractions, we used the bottom-up mean values of tropospheric OH sink (553 Tg/yr), soil oxidation (30 Tg/yr), stratospheric destruction (31 Tg/yr), except tropospheric Cl for three scenarios: 1 Tg/yr ("S20: min trop. Cl"), 11 Tg/yr ("S20 ave."), and 35 Tg/yr ("S20: max trop. Cl"), taken from Table 3 of Saunio et al. (2020). The base KIE scenario is based on Fujita et al. (2020), except stratospheric KIEs derived from balloon sample data collected globally (Röckmann et al., 2011). The same values from the base setting are represented by asterisks. The values of KIEs for the respective sink processes are given at T = 288 K when the temperature dependent KIEs are available (Saueressig et al., 1995, 1996; Gierczak et al., 1997; Tyler et al., 2000).

Table S5. Same as Table 3 in the main text, but for “NoFRESHem” scenario (see Table S3 and Fig. S11). The base results reported in Table 3 (i.e. under the default prior freshwater emissions) is also presented.

	Prior	CH ₄ only	CH ₄ + $\delta^{13}\text{C}$	CH ₄ + δD	CH ₄ + $\Delta^{14}\text{C}$	Base (All)
Fossil Fraction (%)	30.1 [22.5, 37.6]	27.7 [19.3, 36.2]	36.9 [31.2, 42.7]	27.4 [19.7, 35.0]	18.6 [14.1, 23.1]	23.4 [18.9, 27.9]
Error reduction ratio	-	-11%	24%	-1%	41%	41%
Fossil Fraction in Table 3 (%)	23.6 [17.2, 30.1]	23.2 [15.1, 31.6]	31.5 [27.0, 35.9]	23.9 [17.9, 29.8]	18.1 [13.5, 22.7]	23.1 [19.0, 26.8]

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